SALMAN SADIQ

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EDUCATION

M.Sc. Autonomous Systems Sep 2026 - Present
Hochschule Bonn-Rhein-Sieg (H-BRS), Germany

Bachelor of Engineering in Mechatronics Engineering Nov 2021 - Oct 2025
National University of Sciences and Technology (NUST), Pakistan
CGPA: 3.32/4.00
Rector's Gold Medal - Best Final Year Project

TECHNICAL SKILLS AND INTERESTS

Programming:	C, C++, Python, Bash, SQL
Robotics & AI:	ROS2, Gazebo, OpenCV, Computer Vision, VLMs/LLMs, Sensor Integration
Embedded & Communication:	CAN, MQTT, UART, I²C, RTOS, ESP32, Raspberry Pi
Tools:	Linux, Git, Docker, Azure, AWS, KiCad
Languages:	English (fluent), Urdu (native), Punjabi (native), German (A1)

PROFESSIONAL EXPERIENCE

Senior Technical Support Engineer July 2025 - August 2026
Motive Technologies, Inc., Islamabad, PK

- Resolved about 15 technical support tickets daily across software, API, firmware, and backend issues in connected telematics and AI systems using structured root-cause analysis and defect reproduction
- Remotely accessed and diagnosed customer telematics devices using internal tools, communicating directly with customers to resolve issues
- Collaborated with engineering and product teams to reproduce failures, validate fixes, and resolve customer-impacting technical issues

Embedded Design Engineer (Industrial Automation) July 2024 - June 2025
Cowlar Inc. (YC-W17), Islamabad, PK

- Integrated industrial automation systems with client infrastructure, reducing manual processes by 35% and improving operational efficiency by 20%
- Developed hardware/software prototypes using C/C++ and Python, including custom device drivers and automation interfaces, reducing development iteration time by 30%
- Remotely accessed and debugged deployed automation systems over SSH on Linux, diagnosing hardware and software faults at client sites

Embedded Systems Intern Aug 2023 - Sep 2023
RADWI Electronics, Islamabad, PK

- Developed an ESP32-based smart intercom system with audio and camera streaming using I^2C and MQTT
- Developed embedded firmware in C/C++ for device control, sensor data acquisition, and IoT connectivity

PROJECTS

SceneScribe – AI-Enabled Assistive Navigation System

- Developed an AI-powered wearable perception and navigation system for visually impaired users using *Vision-Language Models (VLMs)* and real-time environmental sensing
- Designed a multi-agent VLM architecture for scene understanding, navigation, and contextual assistance; resulting research published at *IEEE IBCAST 2024*

- Integrated camera-based perception with a custom smart cane using *ESP32*, ultrasonic sensing, servo actuation, and haptic feedback
- Designed and prototyped the wearable and smart-cane hardware using CAD and 3D printing; utility patent filed for the adaptive AI-enabled navigation framework

ROS2 Mobile Robot Obstacle Avoidance ROS2, Python, Gazebo

- Developed a ROS2 mobile robot controller in Python, simulated in Gazebo, for reactive LiDAR-based obstacle avoidance
- Processed LiDAR */scan* data to detect obstacles and published velocity commands to */cmd_vel* for autonomous navigation
- Debugged node and topic communication using ROS2 CLI tools such as ros2 topic, ros2 node, and message inspection, mirroring real-world robot troubleshooting

Autonomous Formula Vehicle NUST Automotive Group

- Led a 50-member autonomous vehicle team working across perception, planning, control, and system integration for a Formula Student vehicle
- Led system integration efforts involving CAN, LiDAR/stereo sensing, and path identification across perception and control subsystems
- Led technical training and mentored 20+ members on autonomous vehicle architecture, subsystem integration, and development workflows

Optical Fiber Termination Line Robot @Cowlar

- Developed fiber ferrule-insertion logic and motion control for X, Y, and Z axes using *Festo servo motors* and stepper motors
- Developed and improved *Python* scripts for rapid prototyping, system monitoring, and industrial automation
- Developed embedded firmware and a custom camera interface for real-time logging of production data and inspection images to a web-based monitoring platform
- Implemented a custom communication driver to interface the motor controller with Python over *UART*

CloudSeg – Cloud-Based Image Segmentation App

- Engineered and deployed a *RESTful ML* inference service on an Azure Virtual Machine for image segmentation
- Built a *FastAPI* backend for image processing and *DeepLabV3* inference, returning segmentation masks through REST APIs
- Integrated PostgreSQL with *SQLAlchemy* to track inference jobs, model metadata, and processing status
- Optimized CPU-only inference to approximately 2 seconds per image and implemented secure database access using SSH tunneling

RESEARCH & PATENTS

Z Mahboob, S Sadiq, U Shahbaz Khan, A Zeb, M Islam Tiwana, "Multi-Agent Situational Awareness for Visually Impaired," *2024 International Bhurban Conference on Applied Sciences and Technology (IBCAST), May 2024*
DOI: 10.1109/IBCAST61650.2024.10877205

S Sadiq, Z Mahboob, U Shahbaz Khan, A Zeb, M Islam Tiwana, "Adaptive AI-enabled Framework for Navigation of Visually Impaired Persons," (Patent Filed)

LEADERSHIP & ACHIEVEMENTS

- **General Secretary, SPCS&M, NUST CEME** (May 2024 – May 2025) – Led six student clubs serving 500+ students and received the Medal of Excellence for institutional leadership.
- **2nd Runner Up, IoT & Digital Systems** – COMPPEC.
- **Millennium Fellowship** – Class of 2023.